TEXORA · LECTURE SHEET

Costing of Grey Fabric

A grey fabric's price is its specification, costed line by line: the warp and weft that Sheet 5 weighed, the size that Sheet 3 priced, the picks that Sheet 4 counted, plus the overheads no formula derives. This sheet builds the full cost per metre of the course's reference cloth and reads the result the way a mill manager does.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 13 · SPECIALIZATION — FABRIC MANUFACTURING

1. The cloth being costed

Sheet 5's reference fabric: plain weave, EPI 100, PPI 80, warp and weft Ne 30, GSM 149, grey width 1.5 m — warp 125.6 g and weft 97.6 g per linear metre. Yarn prices (mill's purchase rates): warp Ne 30 at Tk 320/kg, weft Ne 30 at Tk 310/kg.

2. The cost build-up

COST ITEM	BASIS	TK PER METRE	SHARE
Warp yarn	0.1256 kg × Tk 320	40.18	39.3%
Weft yarn	0.0976 kg × Tk 310	30.27	29.6%
Size material	Add-on 10% of warp = 12.56 g solids × Tk 159/kg (Sheet 3)	2.00	2.0%
Sizing process	Labour, steam, power per metre	3.00	2.9%
Weaving charge	3,150 picks/m × Tk 6 per 1,000 picks	18.90	18.5%
Overheads	Depreciation, supervision, repairs, lighting per metre	8.00	7.8%
Total cost		**102.35 ≈ Tk 102.3**	100%

WORKED EXAMPLE — FROM COST TO PRICE

Cost per metre = 40.18 + 30.27 + 2.00 + 3.00 + 18.90 + 8.00 = **Tk 102.3 per metre**.
Cost per m² = 102.3 ÷ 1.5 = **Tk 68.2**.
With a 10% margin: selling price = 102.3 × 1.10 = **Tk 113 per metre**.
Every input is traceable: change any sheet's number — a crimp, a yarn price, a pick rate — and the cost moves by a calculable amount. That traceability, not the total, is what makes this a costing and not a quotation.

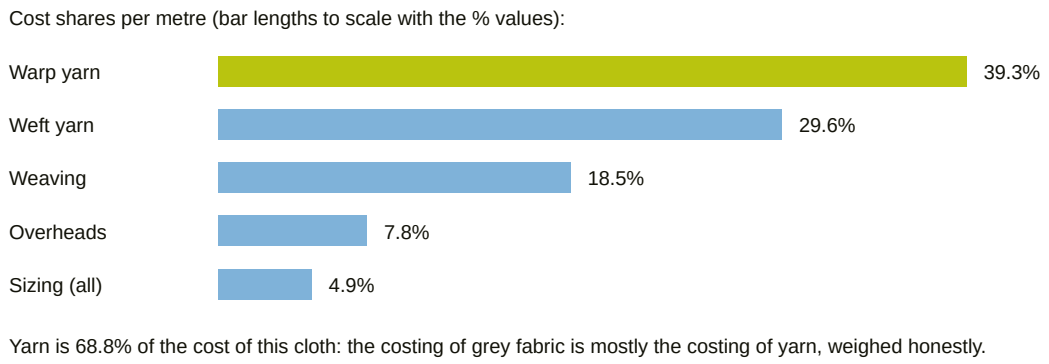


Figure 1 — Where the taka goes: cost shares of the reference fabric.

3. Reading the build-up like a manager

- **Yarn dominates (68.8%).** A 5% rise in yarn price moves the fabric cost by 3.4% — more than the entire sizing bill. Fabric margins are negotiated in the yarn market before they are earned in the shed.
- **The weaving charge is a pick charge.** Costed per 1,000 picks, it makes PPI a price variable: +10 PPI on this cloth adds 394 picks/m and Tk 2.36/m. A buyer's "small" density change is never free.
- **Sizing is small but not negligible** (4.9% all-in), and it is the only item that buys loom performance: cutting it to save 1% of cost can cost 5% of efficiency (Sheet 4, Section 3) — the build-up must be read as a system, not a menu.
- **Overheads are allocated, not measured.** Tk 8.00/m is this mill's rate for this period; the costing is only as honest as the allocation behind that one line.

4. Cost per metre versus cost per kilogram

Buyers of heavy fabrics often quote per kg: here, 223 g/m makes the cost Tk 102.3 ÷ 0.223 = **Tk 459 per kg**. The two prices describe the same cloth; confusing them in a negotiation is a classic and expensive error, which is why every costing sheet states its unit in its title line — per metre, per m², or per kg — before its total.

Glossary

Costing — building a fabric's cost from its specification and current rates. **Weaving charge** — conversion cost of weaving, here rated per 1,000 picks. **Overheads** — period costs allocated to production (depreciation, supervision, repairs). **Margin** — amount added to cost to reach selling price. **Conversion cost** — all costs except the raw material itself.

Key points

• Build-up: warp 40.18 + weft 30.27 + size 2.00 + sizing 3.00 + weaving 18.90 + overheads 8.00 = **Tk 102.3/m** (Tk 68.2/m²; Tk 459/kg; Tk 113/m at 10% margin).
• Yarn is 68.8% of cost — fabric economics is yarn economics.
• A per-pick weaving charge makes PPI a priced variable (Tk 2.36 per +10 PPI here).
• State the unit before the total: per metre, per m² and per kg are different prices for the same cloth.

Practice

1. Recompute the total cost if the warp yarn price rises to Tk 340/kg.
2. The buyer asks for PPI 90 instead of 80, all else unchanged. Find the new weaving charge and the new total cost (weft weight changes too — use Sheet 5's method: weft becomes 73.2 g/m²).
3. Why does a fall in sizing add-on *not* reduce total cost taka-for-taka in practice?

Answer key

1. Warp = 0.1256 × 340 = Tk 42.70; total = 102.35 − 40.18 + 42.70 = **Tk 104.9 per metre**.
2. Picks/m = 90 × 39.37 = 3,543; weaving = 3.543 × 6 = Tk 21.26. Weft per metre = 73.2 × 1.5 = 109.8 g → 0.1098 × 310 = Tk 34.04. Total = 40.18 + 34.04 + 2.00 + 3.00 + 21.26 + 8.00 = **Tk 108.5 per metre** — a 6.1% cost rise for a 12.5% pick rise, because picks are paid for twice: once as yarn, once as weaving.
3. Because add-on is also loom performance: below the optimum, end-breaks rise, efficiency falls (Sheet 4), and the weaving charge per good metre rises faster than the size bill falls. The costing minimum and the size minimum are at different add-ons; the mill optimises the total, not the line.